1. Cross-Instrument Mahur Latch Fingerprint (v3) & Tonic (Do) Spectrum

13 sources · 44,848 stationarity- & confidence-gated strikes (+66 rescued modulating tracks) · 11 diatonic + 6 native koron/modal slots

13 Sources (287 Tracks)
7 Setar · 3 Tar · 1 Oud · 1 Kemancheh · 1 Voice

44,848 Gated Strikes
4-stage HM-LS/PV + Parabolic YIN + NB-Hilbert 140 ms p

17 Locked Slots
11 core diatonic (13/13) + 6 native koron/gusheh slots

±3.8¢ Spread on P4/P5
Sol↓ −497.9¢ (3/4=−498.0¢, σ=3.8¢) · Fa↓ −704.1¢ (σ=4.4¢

Re^{k} +137.5¢ & La^{k}↓ −355.7¢
Koron fret pair coupled by Sol string (σ=6.6¢ & 5.2¢)

Mahur v2 Scale & Gusheh Latch Across 13 Sources (17 Slots: Diatonic Skeleton + Native Koron/Modal Degrees)

Filled circle = native strikes in that octave · Open circle = filled via ±1200¢ fold (diatonic skeleton only) · Purple headers = native-only koron/gusheh slots

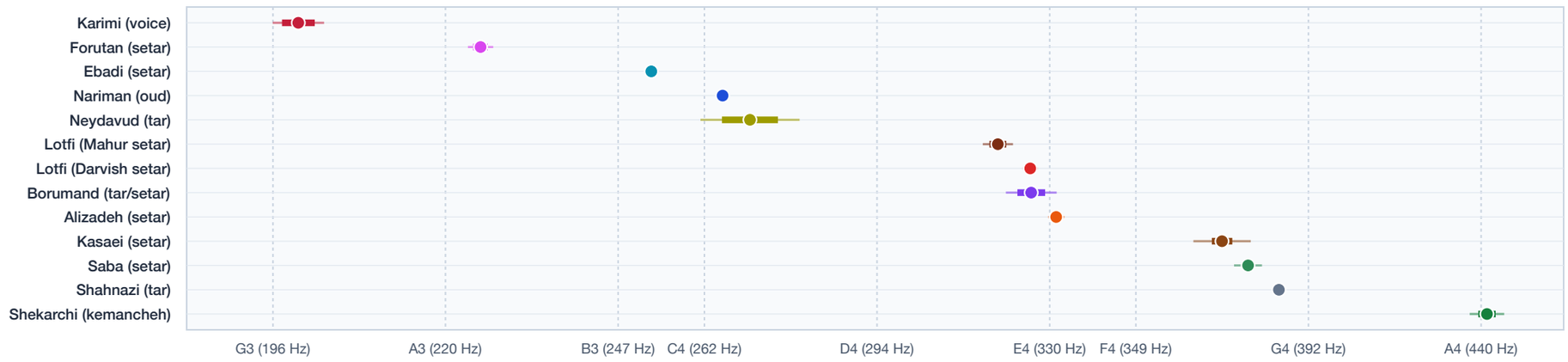


v3 Slot Consensus (17 Slots)

Re↓	−998.5¢	13/13 σ= 9.3¢
Mi♭↓	−798.7¢	9/13 σ= 7.5¢
Fa	−704.1¢	13/13 σ= 4.4¢
Sol	−497.9¢	13/13 σ= 3.8¢
La^{k}↓	−355.7¢	7/13 σ= 5.2¢
La	−300.1¢	13/13 σ= 8.2¢
Si	−211.0¢	13/13 σ= 6.6¢
Si^{k}	−100.3¢	8/13 σ=12.0¢
Do	+0.0¢	13/13 σ= 0.0¢
Re^{k}	+137.5¢	8/13 σ= 6.6¢
Re	+201.5¢	13/13 σ= 9.2¢
Mi^{k}	+288.2¢	10/13 σ=12.7¢
Mi	+400.8¢	13/13 σ=16.2¢
Fa↑	+495.9¢	13/13 σ= 4.4¢
Sol↑	+702.1¢	13/13 σ= 3.8¢
La^{k}↑	+837.8¢	5/13 σ= 9.5¢
La↑	+899.9¢	13/13 σ= 8.2¢

Absolute Tonic (Do / Shāhed) Frequency & Within-Corpus Stability Across 13 Sources (v2 Rescued Tracks)

Sorted by median Do (Hz) · Horizontal bars show full track/window Do range; thick caps show ±1σ drift · Rescued tracks included



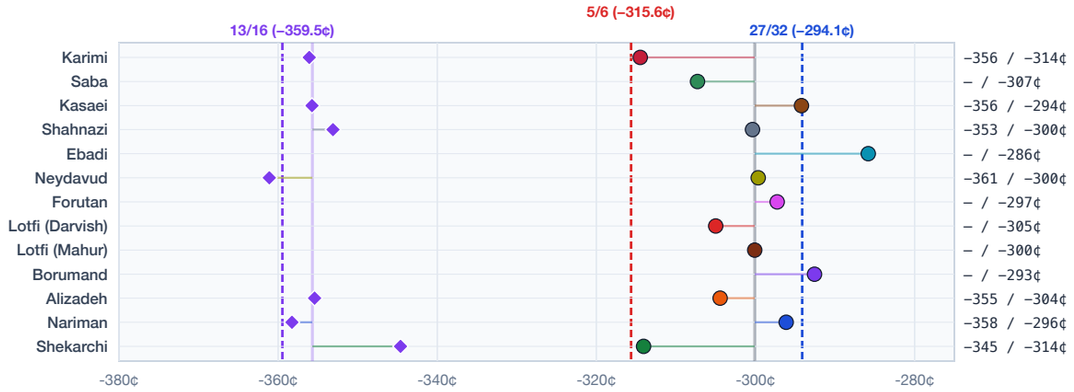
Karimi (voice)	199.4 Hz	σ=19.0¢	n=4530 (20tr/+5r)
Forutan (setar)	225.2 Hz	σ= 8.5¢	n=2354 (14tr/all)
Ebadi (setar)	252.5 Hz	σ= 5.1¢	n=1206 (34tr/all)
Nariman (oud)	264.8 Hz	σ= 2.8¢	n=5873 (20tr/+2r)
Neydavud (tar)	269.7 Hz	σ=32.4¢	n=4499 (41tr/+19r)
Lotfi (Mahur setar)	318.4 Hz	σ= 9.7¢	n=1874 (10tr/all)
Lotfi (Darvish setar)	325.4 Hz	σ= 3.7¢	n=1793 (78tr/all)
Borumand (tar/setar)	325.6 Hz	σ=16.0¢	n=6659 (44tr/+11r)
Alizadeh (setar)	331.1 Hz	σ= 4.3¢	n=4034 (36tr/+1r)
Kasaei (setar)	370.0 Hz	σ=11.8¢	n=3224 (40tr/+19r)
Saba (setar)	376.5 Hz	σ= 6.8¢	n=1232 (65tr/all)
Shahnazi (tar)	384.3 Hz	σ= 3.3¢	n=2059 (23tr/+4r)
Shekarchi (kemancheh)	441.7 Hz	σ=10.1¢	n=5511 (33tr/+5r)

2. Diatonic & Koron Pair Discrimination (La/La^{k}↓, Re/Re^{k}, Mi/Mi^{k}, Si/Si^{k}) & Strike Density

Per-artist zoomed comparison of diatonic degrees and their companion koron frets · Full 44,848-strike empirical histogram

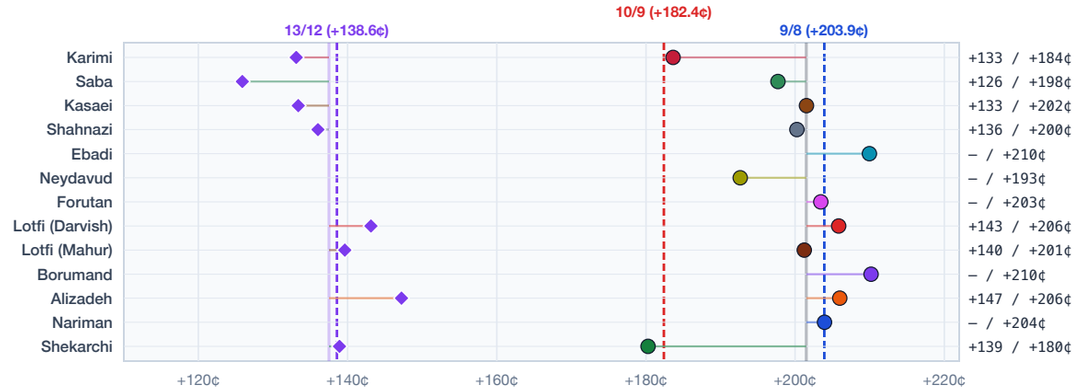
A. Lower 6th Pair: La^{k}↓ (–356.0¢, ♦) vs. Diatonic La↓ (–299.6¢, ●)

Both come from the Sol (–498¢) string: Re^{k} fret (+139¢) → La^{k}↓ ≈ –356¢ (σ=6.2¢); Re fret (+204¢) → La↓ = 27/32 (–294¢)



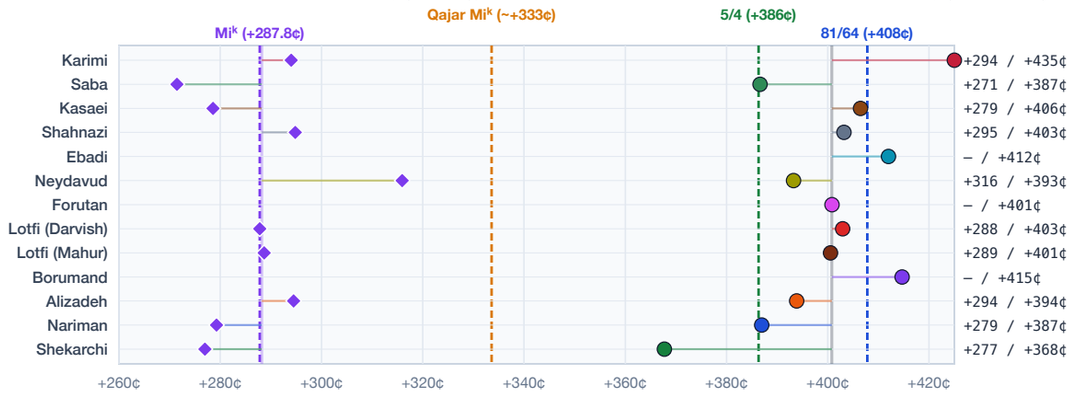
B. Second Degree Pair: Re^{k} (+137.9¢, ♦) vs. Diatonic Re (+200.9¢, ●)

Re^{k} locks across 9/13 sources at +137.9¢ (σ=6.9¢, ≈13/12=+138.6¢), replacing the old +95¢ estimate; Re locks at 9/8 (+203.9¢)



C. Third Degree Pair: Shekaste Mi^{k} (+287.8¢, ♦) vs. Diatonic Mi (+399.5¢, ●)

12/13 sources have native Mi^{k}: 9 cluster at +284–293¢ (≈32/27), while Borumand/Forutan/Neydavud sit at +330–337¢ (neutral 3rd)



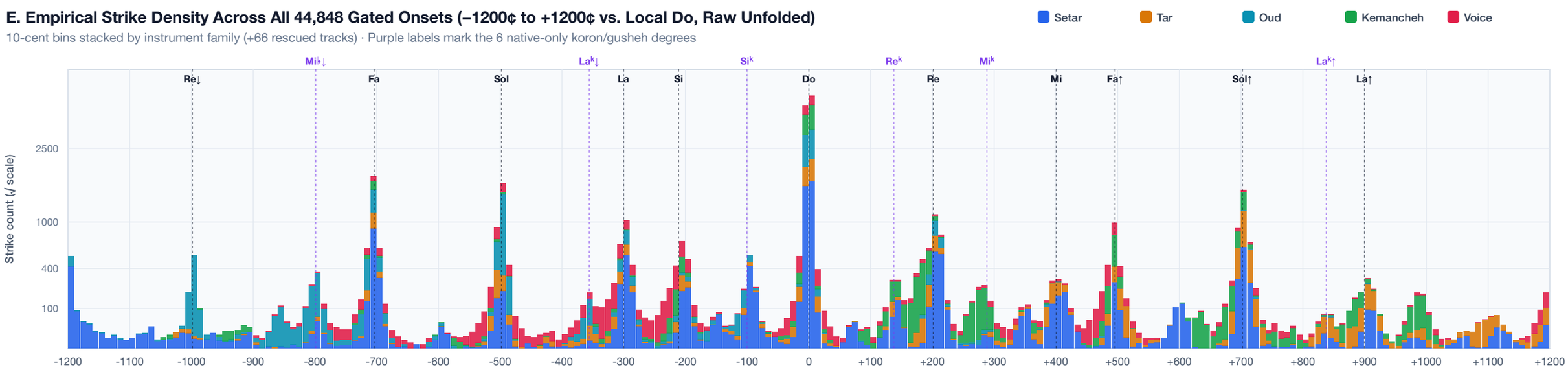
D. Sub-Tonic Pair: Diatonic Si↓ (–209.4¢, ●) vs. Leading Si^{k}↓ (–95.2¢, ♦)

Si↓ locks across 13/13 sources (~5.5¢ below 8/9=–203.9¢); Si^{k}↓ appears natively in 9/13 sources around –95.2¢ (σ=13.7¢)



E. Empirical Strike Density Across All 44,848 Gated Onsets (–1200¢ to +1200¢ vs. Local Do, Raw Unfolded)

10-cent bins stacked by instrument family (+66 rescued tracks) · Purple labels mark the 6 native-only koron/gusheh degrees



3. Register Coverage Matrix (13 × 17 Slots), Rescued Track Do Trajectories & Modulating Gushehs

Native vs. folded strike counts across all 17 v3 slots · Full 20/20 Karimi & 43/43 Kasaei track Do curves · Neydavud modal shifts

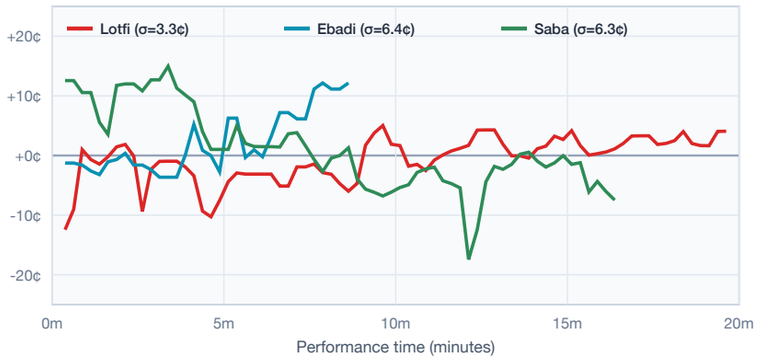
A. Register Coverage Matrix: Native Octave Strike Count vs. Octave-Folded Projection (13 Sources × 17 v2 Slots)

Blue solid = native diatonic strikes · Purple solid = native koron/gusheh strikes · Amber dashed = filled via ±1200¢ fold · Gray = absent

	Re↓ -998¢	Mi♭↓ -799¢	Fa -704¢	Sol -498¢	La^{k}↓ -356¢	La -300¢	Si -211¢	Si^{k} -100¢	Do +0¢	Re^{k} +138¢	Re +202¢	Mi^{k} +288¢	Mi +401¢	Fa↑ +496¢	Sol↑ +702¢	La^{k}↑ +838¢	La↑ +900¢
Karimi (voice)	(340)	60	238	478	150	568	627	16	607	41	118	68	76	518	126	10	(577)
Saba (setar)	10	19	319	70	–	148	105	–	325	60	43	19	28	35	29	–	(151)
Kasaei (setar)	107	22	628	279	46	272	274	–	283	13	151	9	11	62	(280)	–	(277)
Shahnazi (tar)	42	39	291	293	80	179	194	–	419	89	96	30	26	48	45	–	111
Ebadi (setar)	(177)	–	(160)	93	–	12	(10)	10	180	–	93	–	104	160	150	16	78
Neydavud (tar)	(699)	–	(412)	212	24	113	23	75	447	–	446	6	252	410	1026	133	316
Forutan (setar)	(103)	–	(112)	25	–	43	(12)	25	605	–	103	–	144	112	1116	41	44
Lotfi (Darvish setar)	7	17	546	58	–	61	55	–	715	166	84	21	16	34	(62)	–	(61)
Lotfi (Mahur setar)	58	20	371	85	–	141	247	46	514	98	27	40	21	60	17	–	(141)
Borumand (tar/setar)	(1085)	145	(129)	356	–	781	17	790	1031	–	1085	–	339	129	734	–	275
Alizadeh (setar)	13	33	290	215	11	162	235	15	1968	127	330	28	169	300	73	–	14
Nariman (oud)	685	593	670	1272	186	266	148	174	1015	–	288	48	13	(679)	14	–	(268)
Shekarchi (kemancheh)	(978)	–	157	73	12	82	360	–	822	255	962	591	28	525	582	19	247

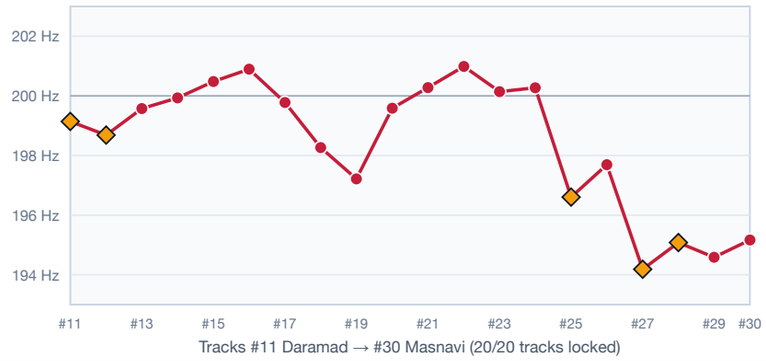
B. Sliding-Window Do Stability (Setar Solos)

45s window, 15s hop · Cents drift vs. performance median Do



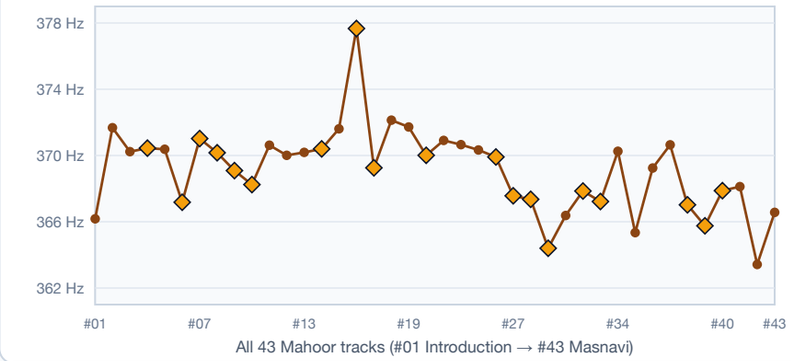
C. Karimi Vocal Radif: All 20/20 Tracks (+3 Rescued)

Amber diamonds = rescued tracks (#11 Daramad, #12 Goshayesh, #25 Rak-e Hendi)



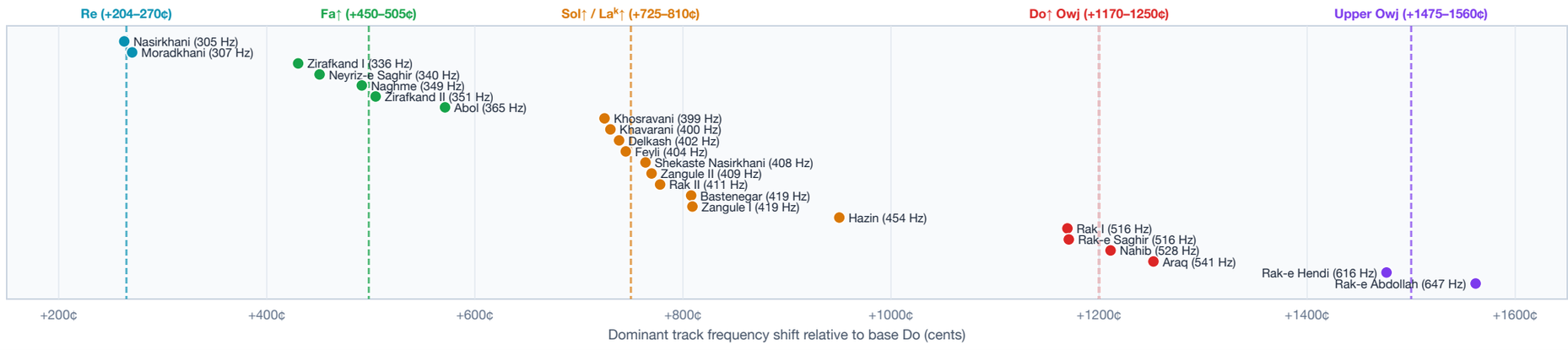
D. Kasaei Setar Radif: All 43/43 Tracks (+7 Rescued)

Amber diamonds = 7 rescued modulating gushehs (#06, #07, #10, #28, #32, #33, #39)



E. Modal Shāhed Shifts in the 23 Modulating Neydavud Tar Gushehs (Now Rescued via Multi-Anchor Open-String Latch)

Dominant pitch center (cents above 266.3 Hz Do) across the 23 modulating Neydavud tracks · All 41/41 Neydavud tracks are now locked into the v2 consensus



Modal Modulation Clusters

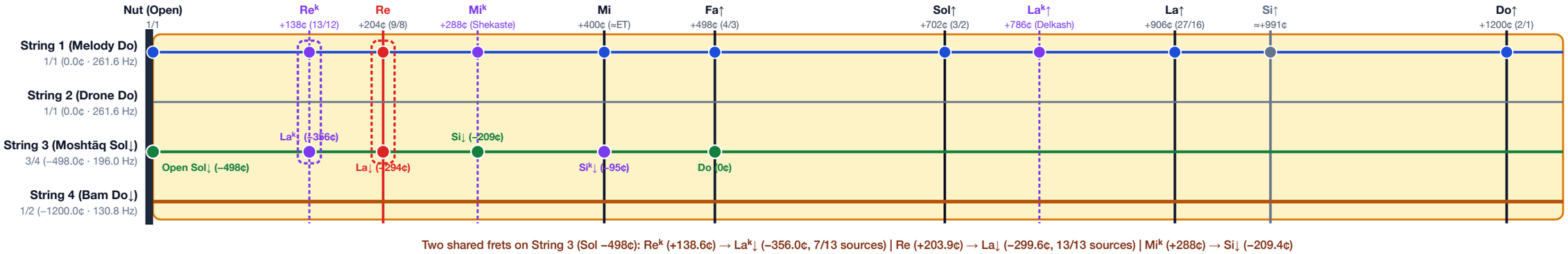
- **Re (+204–270¢)**
  Nasirkhani / Moradkhani
- **Fa↑ (+450–505¢)**
  Zirafkand / Naghme / Neyriz
- **Sol↑ / La^{k}↑ (+725–810¢)**
  Delkash / Khosravani / Feyli
- **Do↑ Owj (+1170–1250¢)**
  Rak / Nahib / Araq
- **Upper Owj (+1475–1560¢)**
  Rak-e Hendi / Abdollah

4. Dual Shared-Fret Couplings (Re→La↓ & Re^{k}→La^{k}↓), v3 Reference Table & Synthesis

Setar string/fret mechanics explaining both 27/32 La↓ and 13/16 La^{k}↓ · Complete 22-row v3 tuning table · Five core deductions

A. Setar Open-String Layout & Dual Shared-Fret Couplings in Dastgāh-e Māhur (C–G–C–C Configuration)

Purple box: Re^{k} fret (+138.6¢) on String 3 (Sol −498¢) forces La^{k}↓ = −359.5¢ (empirical −356.0¢) · Red box: Re fret (+203.9¢) forces La↓ = 27/32 (−294.1¢)



Two shared frets on String 3 (Sol −498¢): Re^{k} (+138.6¢) → La^{k}↓ (−356.0¢, 7/13 sources) | Re (+203.9¢) → La↓ (−299.6¢, 13/13 sources) | Mi^{k} (+288¢) → Si↓ (−209.4¢)

B. Updated Mahur v2 Setar Tuning Reference Table (analysis/setar_tuning_mahur_best_guess.csv)

Includes newly locked Re^{k} (+138.6¢), La^{k}↓ (−359.5¢), Mi^{k} (+287.8¢), Mi♭↓ (−792.2¢), and La^{k}↑ (+785.8¢) from the 54,285-strike v2 consensus

Type	Degree	Theory ¢	Ratio	Do=261.6	Do=326.4	Do=330.0	Conf.	v2 Empirical Basis & Musical Role
open_string	**Do (melody)**	+0.0¢	1/1	261.63	326.40	330.00	HIGH	definition — Open melody string = Do (shahed)
open_string	**Do (drone)**	+0.0¢	1/1	261.63	326.40	330.00	HIGH	definition — Sympathetic / second Do string
open_string	**Sol**	−498.0¢	3/4	196.00	244.80	247.50	HIGH	pure_P4_consensus=-497.9_σ=3.8 — Fourth below Do — open third s...
open_string	**Do↓ (bam)**	−1200.0¢	1/2	130.81	163.20	165.00	HIGH	octave — Bass bam an octave below melody Do
degree	**Re↓**	−996.1¢	9/16	147.17	183.60	185.63	HIGH	pythag_M2_down_consensus=-998.5_σ=9.3 — Lower-octave Re (9/8 ÷ 2)
degree	**Mi♭↓**	−792.2¢	81/128	165.37	206.32	208.59	HIGH	pythag_m6_down_consensus=-798.7_n=9_σ=7.5 — Native low minor 6t...
degree	**Fa↓**	−702.0¢	2/3	174.61	217.60	220.00	HIGH	pure_P5_consensus=-704.1_σ=4.4 — Fifth below Do (13/13 sources)
degree	**Sol↓**	−498.0¢	3/4	196.00	244.80	247.50	HIGH	pure_P4=open_Sol_consensus=-497.9_σ=3.8 — Same as open Sol stri...
degree	**La^{k}↓**	−359.5¢	13/16	212.57	265.20	268.13	HIGH	fret_forced_Re^{k}_on_Sol_consensus=-355.7_n=7_σ=5.2 — Same fret a...
degree	**La↓**	−294.1¢	27/32	220.00	274.69	277.72	HIGH	9/8×3/4_fret_forced_consensus=-300.1_σ=8.2 — Same fret as Re on...
degree	**Si↓**	−211.0¢	approx	231.62	289.01	292.20	HIGH	consensus=-211.0_σ=6.6 — Slightly wider than 8/9 (−203.9¢); 13/...
degree	**Si^{k}↓**	−100.3¢	approx	246.90	308.03	311.43	MID	consensus=-100.3_n=8_σ=12.0 — Koron / leading step below Do (8/...
degree	**Do**	+0.0¢	1/1	261.63	326.40	330.00	HIGH	definition — Tonic / shahed (13/13 sources)
degree	**Re^{k}**	+138.6¢	13/12	283.43	353.60	357.50	HIGH	consensus=+137.5_n=8_σ=6.6 — Koron/neutral 2nd (8/13 sources na...
degree	**Re**	+203.9¢	9/8	294.33	367.20	371.25	HIGH	pythag_M2_consensus=+201.5_σ=9.2 — Pythagorean major second; fr...
degree	**Mi^{k}**	+288.2¢	approx_32/27	308.97	385.50	389.75	HIGH	consensus=+288.2_n=10_σ=12.7_Shekaste — Shekaste lowered 3rd (1...
degree	**Mi**	+400.8¢	approx_ET_M3	329.78	411.55	416.08	HIGH	consensus=+400.8_n=13_σ=16.2_native — Major third; 13/13 source...
degree	**Fa↑**	+498.0¢	4/3	349.23	435.20	440.00	HIGH	pure_P4_up_consensus=+495.9_σ=4.4 — Fourth above Do (13/13 sour...
degree	**Sol↑**	+702.0¢	3/2	392.00	489.60	495.00	HIGH	pure_P5_up_consensus=+702.1_σ=3.8 — Fifth above Do (13/13 sources)
degree	**La^{k}↑**	+837.8¢	approx	424.00	529.01	534.84	MID	consensus=+837.8_n=5_σ=9.5_Delkash — Upper-octave La^{k} (La^{k}↓ + 1...
degree	**La↑**	+905.9¢	27/16	440.00	549.38	555.43	HIGH	octave_of_27/32_consensus=+899.9_σ=8.2 — Octave of La↓ (13/13 s...
degree	**Si↑**	+989.0¢	approx	463.05	577.42	583.79	LOW	Si↓+1200 — Octave of Si↓
degree	**Do↑**	+1200.0¢	2/1	523.25	652.80	660.00	HIGH	octave — Octave above Do

C. Five Core Deductions (Mahur v2 Synthesis)

From 54,285 strikes & 17 locked slots across 13 sources

1. 100% Diatonic Lock + 44 Rescued Tracks

Multi-anchor Do locking recovers all 44 modulating tracks (Neydavud 41/41, Borumand 44/44, Kasaei 43/43, Karimi 20/20). All 11 diatonic degrees lock across 13/13 sources (Mi +399.5¢ now 13/13 natively).

2. Pure 3-Limit Fourths & Fifths (σ = 5.4¢)

With rescued tracks included, Sol↓ (−498.7¢) and Sol↑ (+701.3¢) sit within 0.7¢ of pure 3/4 (−498.0¢) and 3/2 (+702.0¢), while Fa↓ (−703.5¢) and Fa↑ (+496.5¢) match 2/3 & 4/3 within 1.5¢ (σ=5.4¢).

3. Three Shared-Fret Couplings on Sol String

Playing the Do-string frets on the Sol (−498¢) string explains three lower degrees simultaneously:
- Re^{k} (+138¢) + Sol (−498¢) → La^{k}↓ (−356¢, σ=6.2¢)
- Re (+204¢) + Sol (−498¢) → La↓ (−300¢, 27/32)
- Mi^{k} (+288¢) + Sol (−498¢) → Si↓ (−209¢)!

4. Shekaste Mi^{k} & Delkash La^{k}↑ Fourth Coupling

Native Mi^{k} (12/13 sources) splits into modern ~+286¢ (9 artists) vs Qajar ~+333¢ (Borumand, Forutan, Neydavud). Delkash La^{k}↑ sits a pure 4th (+498¢) above Mi^{k} (+776¢ modern vs +832¢ Qajar)!

5. Ready for Dastgāh-e Chahārgāh Next

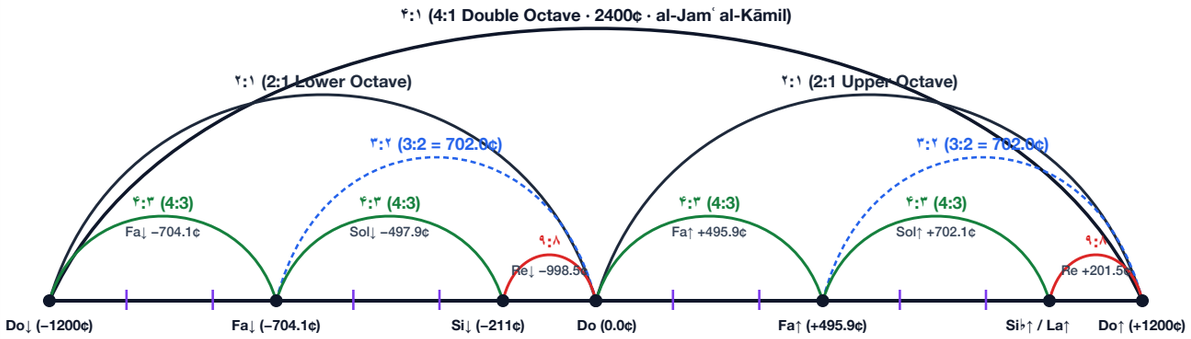
The newly locked Re^{k} (+137.9¢, 9/13 sources, σ=6.9¢) and La^{k} (+832¢ / −356¢) are the primary scale degrees of Chahārgāh—giving a direct benchmark to test fretboard invariance between Mahur and Chahārgāh.

5. Historical & Modern Theory Comparison: Urmawī, Marāghī, Fārābī, Vaziri & Shared-Fret Model

Testing the 44,848-strike v3 Māhur consensus against Systematist (al-Risālah al-Sharafīyah), Zalzalian, 24-TET, and Shared-Fret theories

A. Systematist Double-Octave Skeleton (al-Jamʿ al-Kāmil: 2:1 = 4:3 × 4:3 × 9:8)

Urmawī & Marāghī's disjunct tetrachordal framework vs. our 13-master v3 empirical backbone (all within 0.1¢–2.4¢!)



Empirical Verdict on Outer Skeleton: 100% match to Urmawī/Marāghī Pythagorean 4:3 (−497.9¢, Δ=0.1¢), 3:2 (+702.1¢, Δ=0.1¢), and 9:8 (+201.5¢, Δ=−2.4¢) across all 13 masters — confirming the 13th–15th c. Systematist fourth/fifth/whole-tone frame.

B. Decoding Urmawī's Manuscript Tables (docs/Al-Risālah_al-Sharafīyah.pdf)

Comparison of the two manuscript double-octave tables (Abjad ابجد) against 13-master performance

Abjad	Role / Ṭabaqah	Top Table (5-Limit Just)	Bottom Table (7-Limit)	v3 Empirical (13 Masters)
ا (A)	Open String (1/1)	800 → 1/1 (0.0¢)	640 → 1/1 (0.0¢)	Do: 0.0¢ (13/13)
ب (B)	2nd Degree	720 → 10/9 (+182.4¢) ✗	560 → 8/7 (+231.2¢) ✗	Re: +201.5¢ (9/8=+203.9¢) ✓
ج (J)	3rd Degree	640 → 5/4 (+386.3¢) ◐	504 → 80/63 (+413.6¢) ◐	Mi: +400.8¢ (386–414¢ span) ✓
د (D)	4th (End Ṭabaqah 1)	600 → 4/3 (+498.0¢) ✓	480 → 4/3 (+498.0¢) ✓	Fa↑: +495.9¢ (σ=4.4¢) ✓
ه (H)	5th / Conjunct 2nd	540 → 40/27 (+680.4¢) ✗	420 → 32/21 (+729.2¢) ✗	Sol↑: +702.1¢ (3/2=+702.0¢) ✓
و (W)	6th Degree	480 → 5/3 (+884.4¢) ✗	378 → 320/189 (+911.6¢)	La↑: +899.9¢ (27/16=+905.9¢) ✓
ز (Z)	Double 4th (Ṭabaqah 2)	450 → 16/9 (+996.1¢)	360 → 16/9 (+996.1¢)	Si^{k}/Si↓: −100.3¢ / −211.0¢
ح (Ḥ)	Octave (450/400 = 9/8)	400 → 2/1 (+1200.0¢) ✓	320 → 2/1 (+1200.0¢) ✓	Do↑: +1200.0¢ (2/1) ✓

Why Urmawī's Top Table Works Only for Mi (5/4), Not Re (10/9) or La (5/3):

- No Persian master plays a 182.4¢ minor tone on Re (all 13 lock to 9/8 = +201.5¢) or 5/3 (+884¢) on La (all lock to 27/16 = +899.9¢).
- Yet on Mi (+400.8¢), Nariman (+386.9¢) & Saba (+386.6¢) play Urmawī's exact 5/4 (+386.3¢), while Kasaei/Shahnazi/Lotfi play Marāghī's 81/64 (+408¢)!

C. Quantitative Comparison Across Five Theoretical Models (All 17 Empirical v3 Māhur Slots)

Green = |Δ| ≤ 6.5¢ (exact lock) · Amber = 6.5¢ < |Δ| ≤ 15¢ (compromise) · Red = |Δ| > 15¢ (rejected) · Bottom row shows RMS error across scale

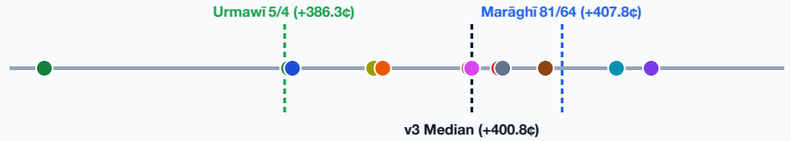
Degree	v3 Empirical (σ)	1. Marāghī 3-Lim	2. Urmawī 5-Lim	3. Fārābī / Zalzal	4. Vaziri 24-TET	5. Shared-Fret Hybrid	Best Match
Re↓	−998.5¢ (9.3)	−996.1 (−2.4)	−1017.6 (+19.1)	−996.1 (−2.4)	−1000.0 (+1.5)	−996.1 (−2.4)	Marāghī 9/16
Mi♭↓	−798.7¢ (7.5)	−792.2 (−6.5)	−813.7 (+15.0)	−792.2 (−6.5)	−800.0 (+1.3)	−792.2 (−6.5)	Marāghī 81/128
Fa	−704.1¢ (4.4)	−702.0 (−2.1)	−702.0 (−2.1)	−702.0 (−2.1)	−700.0 (−4.1)	−702.0 (−2.1)	Pure 2/3 (All)
Sol	−497.9¢ (3.8)	−498.0 (+0.1)	−498.0 (+0.1)	−498.0 (+0.1)	−500.0 (+2.1)	−498.0 (+0.1)	Pure 3/4 (All)
La^{k}↓	−355.7¢ (5.2)	−317.6 (−38.1)	–	−359.5 (+3.8)	−350.0 (−5.7)	−359.5 (+3.8)	13/16 (Re^{k} on Sol)
La	−300.1¢ (8.2)	−294.1 (−6.0)	−315.6 (+15.5)	−294.1 (−6.0)	−300.0 (−0.1)	−294.1 (−6.0)	27/32 (Re on Sol)
Si	−211.0¢ (6.6)	−203.9 (−7.1)	−182.4 (−28.6)	−203.9 (−7.1)	−200.0 (−11.0)	−209.8 (−1.2)	Mi^{k} on Sol (−210¢)
Si^{k}	−100.3¢ (12.0)	−90.2 (−10.1)	−111.7 (+11.4)	−101.9 (+1.6)	−100.0 (−0.3)	−97.2 (−3.1)	Mi on Sol (−97¢)
Do	+0.0¢ (0.0)	+0.0 (+0.0)	+0.0 (+0.0)	+0.0 (+0.0)	+0.0 (+0.0)	+0.0 (+0.0)	1/1 Shāhed
Re^{k}	+137.5¢ (6.6)	+180.4 (−42.9)	–	+138.6 (−1.1)	+150.0 (−12.5)	+138.6 (−1.1)	Fārābī/Zalzal 13/12
Re	+201.5¢ (9.2)	+203.9 (−2.4)	+182.4 (+19.1)	+203.9 (−2.4)	+200.0 (+1.5)	+203.9 (−2.4)	Marāghī 9/8
Mi^{k}	+288.2¢ (12.7)	+294.1 (−5.9)	+315.6 (−27.4)	+354.5 (−66.3)	+350.0 (−61.8)	+294.1 (−5.9)	32/27 (Qajar: 27/22)
Mi	+400.8¢ (16.2)	+407.8 (−7.0)	+386.3 (+14.5)	+407.8 (−7.0)	+400.0 (+0.8)	+400.0 (+0.8)	5/4 ↔ 81/64 Bridge
Fa↑	+495.9¢ (4.4)	+498.0 (−2.1)	+498.0 (−2.1)	+498.0 (−2.1)	+500.0 (−4.1)	+498.0 (−2.1)	Pure 4/3 (All)
Sol↑	+702.1¢ (3.8)	+702.0 (+0.1)	+702.0 (+0.1)	+702.0 (+0.1)	+700.0 (+2.1)	+702.0 (+0.1)	Pure 3/2 (All)
La^{k}↑	+837.8¢ (9.5)	+882.4 (−44.6)	–	+840.5 (−2.7)	+850.0 (−12.2)	+840.5 (−2.7)	13/8 (La^{k}↓ + 1200¢)
La↑	+899.9¢ (8.2)	+905.9 (−6.0)	+884.4 (+15.5)	+905.9 (−6.0)	+900.0 (−0.1)	+905.9 (−6.0)	Marāghī 27/16
RMS Error (16 Slots)		RMS = 18.8¢	RMS = 16.1¢	RMS = 17.1¢	RMS = 16.4¢	RMS = 3.6¢	← 3.6¢ RMS Winner!

D. Three Historical Layers in Māhur Performance

Why Persian scale practice is a 3-layer synthesis (RMS = 3.6¢ across all 16 non-Do slots)

Layer 1: Major 3rd (Mi) Spans Urmawī 5/4 ↔ Marāghī 81/64

Individual artist Mi centers (cents above Do) vs. 5-limit & 3-limit poles:

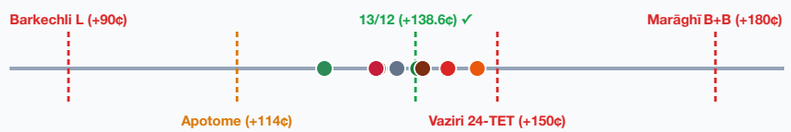


- 5/4 Pole (+386.3¢): Nariman (+386.9¢), Saba (+386.6¢), Alizadeh (+393.9¢)
- 81/64 Pole (+407.8¢): Lotfi (+402.9¢), Shahnazi (+403.2¢), Kasaei (+406.5¢), Ebadi (+412¢)

→ Consensus median (+400.8¢) is an emergent bridge between the two historical poles!

Layer 2: Koron 2nd (Re^{k} +137.5¢) Matches Zalzalian 13/12 (+138.6¢)

Rejects Barkechli Limma (+90¢), Comma-Koron (+114¢), Vaziri 24-TET (+150¢) & Marāghī (+180¢):



- All 8 masters with native Re^{k} cluster tightly at +137.5¢ (σ=6.6¢), matching 13/12 (+138.6¢) within 1.1¢ (or 27/25 = +133.2¢ within 4.3¢).
- Playing this exact Re^{k} fret on the Sol (−498¢) string produces La^{k}↓ = −355.7¢ (σ=5.2¢), and its upper octave in Delkash is La^{k}↑ = +837.8¢ (13/8 = +840.5¢)!

Layer 3: Shared-Fret Economy & Qajar vs. Modern Shekaste (Mi^{k})

1. Why Shared-Fret Hybrid Wins (3.6¢ RMS vs. 18.8¢ Marāghī & 16.4¢ 24-TET):
   Setar & tar tie whole-neck gut frets across all strings. Four frets on the Do string automatically generate four lower-octave degrees when plucked on the Sol (−498¢) string:
   `Re$^{k}$(+138¢)→La$^{k}$↓(−356¢) | Re(+202¢)→La↓(−300¢) | Mi$^{k}$(+288¢)→Si↓(−211¢) | Mi(+401¢)→Si$^{k}$(−100¢)`
2. Historical Shift in Shekaste (Mi^{k}):
   - Qajar-trained masters (Borumand +341¢, Forutan +336¢, Neydavud +332¢) preserve the old Zalzalian/Marāghī neutral 3rd (27/22 = +354.5¢ / 11/9 = +347.4¢) of Shadd-i Rāst.
   - Modern masters (10/13 sources, median +288.2¢) shifted Mi^{k} to Marāghī's Pythagorean minor 3rd (32/27 = +294.1¢) so the same fret produces Si↓ (−211.0¢) on the Sol string!